

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250277322

Kind Code

A1

Publication Date

September 04, 2025

Inventor(s)

Villalon, JR.; Thomas Anthony et al.

ATMOSPHERIC CONTROL OF ENCLOSED ELECTROLYTIC CELLS

Abstract

Methods of performing electrolytic reactions in scaled electrolytic cells, methods of passivating electrolytic cell components, and associated systems and electrolytic cells are generally described. Some methods comprise performing electrolytic reactions in sealed electrolytic cells in a manner that results in the passivation of one or more electrolytic cell components.

Inventors: Villalon, JR.; Thomas Anthony (Boston, MA), Salvucci; Richard Robert (Quincy, MA), Kim; Peace (Reading, MA), Hutcheson; Henry Chase (Malden, MA)

Applicant: Phoenix Tailings, Inc. (Woburn, MA)

Family ID: 95024743

Assignee: Phoenix Tailings, Inc. (Woburn, MA)

Appl. No.: 19/066741

Filed: February 28, 2025

Related U.S. Application Data

us-provisional-application US 63560595 20240301

Publication Classification

Int. Cl.: C25C3/18 (20060101); C25C3/02 (20060101); C25C3/34 (20060101); C25C5/04 (20060101)

U.S. Cl.:

CPC C25C3/18 (20130101); C25C3/02 (20130101); C25C3/34 (20130101); C25C5/04 (20130101);

Background/Summary

RELATED APPLICATIONS [0001] This application claims priority to U.S. Provisional Application No. 63/560,595, filed Mar. 1, 2024, which is incorporated herein by reference in its entirety for all purposes.

TECHNICAL FIELD

[0002] Methods of performing electrolytic reactions in sealed electrolytic cells, methods of passivating electrolytic cell components, and associated systems and electrolytic cells are generally described.

SUMMARY

[0003] Methods of performing electrolytic reactions in sealed electrolytic cells, methods of passivating electrolytic cell components, and associated systems and electrolytic cells are generally described. The subject matter of the present disclosure involves, in some cases, interrelated products, alternative solutions to a particular problem, and/or a plurality of different uses of one or more systems and/or articles.

[0004] Certain embodiments relate to methods of passivating a component of an electrolytic cell. In some embodiments, a method of passivating the component of the electrolytic cell comprises performing an electrolytic reaction in the electrolytic cell, wherein the electrolytic reaction comprises reducing metallic ions to form a metal and comprises forming a byproduct; and depositing the byproduct and/or a reaction product of the byproduct on the component, thereby passivating the component.

[0005] Certain embodiments relate to methods.

[0006] In some embodiments, a method comprises performing an electrolytic reaction in an electrolytic cell, wherein the electrolytic reaction comprises reducing metallic ions to form a metal and forming a byproduct, and wherein, during the electrolytic reaction, the electrolytic cell traps the byproduct and/or a reaction product of the byproduct.

[0007] In some embodiments, a method comprises performing an electrolytic reaction in an electrolytic cell, wherein: the electrolytic reaction comprises reducing metallic ions to form a metal, during the electrolytic reaction, the electrolytic cell is contained in a container, the container is fluidically connected to an environment, the environment comprises a first species reactive with a second species present in the electrolytic cell during the electrolytic reaction, and during the electrolytic reaction, the container is sealed with respect to the first species.

[0008] In some embodiments, a method comprises performing an electrolytic reaction in the electrolytic cell, wherein the electrolytic reaction comprises reducing metallic ions to form a metal; and during the electrolytic reaction, depositing a species present in the electrolytic cell on a component of the electrolytic cell, thereby passivating the component.

[0009] In some embodiments, a method comprises depositing a species present in an electrolytic cell on a component of the electrolytic cell, thereby passivating the component.

[0010] Certain embodiments relate to systems. In some embodiments, a system comprises an electrolytic cell; and a container containing the electrolytic cell, wherein the container comprises a

plurality of reversibly sealable ports.

[0011] Other advantages and novel features of the present disclosure will become apparent from the following detailed description of various non-limiting embodiments of the disclosure when considered in conjunction with the accompanying figures. In cases where the present specification and a document incorporated by reference include conflicting and/or inconsistent disclosure, the present specification shall control.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0012] Non-limiting embodiments of the present disclosure will be described by way of example with reference to the accompanying figures, which are schematic and are not intended to be drawn to scale unless otherwise indicated. In the figures, each identical or nearly identical component illustrated is typically represented by a single numeral. For purposes of clarity, not every component is labeled in every figure, nor is every component of each embodiment of the disclosure shown where illustration is not necessary to allow those of ordinary skill in the art to understand the disclosure. In the figures:

[0013] FIG. 1 is a schematic illustration of a system comprising an electrolytic cell and a container containing the electrolytic cell, in accordance with certain embodiments.

[0014] FIG. 2 is a schematic illustration of a system comprising an electrolytic cell comprising an anode and a cathode and contained within a container, in accordance with certain embodiments.

[0015] FIG. 3 is a schematic illustration of a system comprising an electrolytic cell that comprises an electrolyte in which the electrodes are partially immersed, in accordance with some embodiments.

[0016] FIG. 4 is a schematic illustration of a system comprising an electrolytic cell that comprises an electrolyte container, in accordance with some embodiments.

[0017] FIG. 5 is a schematic illustration of a system comprising a container comprising a plurality of reversibly sealable ports, in accordance with some embodiments.

[0018] FIG. 6 is a schematic illustration of a container fluidically connected to an environment, in accordance with some embodiments.

[0019] FIGS. 7 and 8 are flow diagrams of methods, in accordance with certain embodiments.

[0020] FIG. 9 is a schematic illustration of an exemplary system, in accordance with certain embodiments.

DETAILED DESCRIPTION

[0021] Methods of performing electrolytic reactions in sealed electrolytic cells, methods of passivating electrolytic cell components, and associated systems and electrolytic cells are generally described. Some methods comprise performing electrolytic reactions in sealed electrolytic cells in a manner that results in the passivation of one or more electrolytic cell components.

[0022] In some embodiments, an electrolytic reaction may be performed in an electrolytic cell in a manner that is particularly beneficial. For instance, in some embodiments, an electrolytic reaction may be performed in an electrolytic cell in a manner that results in the formation of a byproduct that causes the passivation of a component of the electrolytic cell. The byproduct itself may directly passivate the relevant component and/or may undergo one or more further reactions that result in a species that passivates the relevant component. The formation of such byproducts may be desirable because they may result in the electrolytic cell becoming less reactive with one or more species contained therein over time (e.g., over the course of an electrolytic reaction, over the course of multiple electrolytic reactions and/or uses thereof), rendering it increasingly suitable for performing electrolytic reactions and/or allowing electrolytic reactions to be performed therein with fewer deleterious side reactions. Such species may include species initially present in the electrolytic cell

(e.g., in the electrolyte, in an environment contained by a container in which the electrolytic cell is positioned) and/or generated during an electrolytic reaction performed in the electrolytic cell (e.g., products, byproducts, and/or reaction products thereof generated during the performance of an electrolytic reaction).

[0023] It is also possible for an electrolytic reaction to be performed in an electrolytic cell that results in the passivation of a component thereof due to a species that is not a byproduct or a reaction product of a byproduct. As one example, in some embodiments, an electrolytic cell comprises one or more gases (e.g., in trace amounts, that are contaminants, and/or that are introduced thereinto concurrently with the introduction of the material undergoing the electrolytic reaction) that passivate a component thereof.

[0024] In some embodiments, a byproduct, a reaction product of a byproduct, and/or a species that is neither of the foregoing passivates a component of an electrolytic cell via the Boudouard reaction. Performing passivation in this manner may desirably also remove carbon dust from electrolytic cells and/or deposit a carbon-containing passivating layer comprising carbon originating from carbon dust.

[0025] As another example, in some embodiments, an electrolytic reaction is performed in an electrolytic cell that traps one or more byproducts formed therein. Such byproducts may have one or more advantageous effects, such as the direct and/or indirect passivation of an electrolytic cell component as described above. Trapping such byproducts in the electrolytic cell may advantageously result in a higher efficiency and/or larger extent of the advantageous effect(s), such as a higher efficiency of passivation and/or more rapid passivation of an electrolytic cell component by the byproduct.

[0026] As a third example, in some embodiments, an electrolytic reaction is performed in an electrolytic cell that is sealed with respect to one or more species present external thereto that it would be undesirable to introduce thereinto. For instance, an electrolytic cell may be positioned in an environment comprising a species that is reactive with a species present in the electrolytic cell and/or reactive with a species generated during the electrolytic reaction. When the electrolytic cell is sealed with respect to the reactive species present in the external environment, deleterious reactions between that species and the species present in the electrolytic cell with which it is reactive may desirably be reduced appreciably and/or prevented. This may reduce and/or prevent unwanted consumption of species produced and/or generated in such electrolytic cells via reactions with the species in the external environment. In some embodiments, such sealing may reduce and/or prevent the generation of undesirable species inside the electrolytic cell via reaction between the species to which the electrolytic cell is sealed and species inside the electrolytic cell.

[0027] Some embodiments relate to systems. Such systems may be capable of performing and/or configured to perform some or all of the steps of the methods described herein. Similarly, some methods may be performed in the systems described herein. In some embodiments, a system has one or more features that promote the passivation of components therein and/or that assist with sealing. As one example, in some embodiments, a system comprises a plurality of reversibly scalable ports. Such ports may be suitable for both allowing the introduction and/or removal of one or more species into and/or out of the electrolytic cell when desired (e.g., prior to and/or after the performance of an electrolytic reaction) and sealing the electrolytic (e.g., during an electrolytic reaction). Some such ports may allow for the introduction of one or more species into and/or the removal of one or more species out of the electrolytic cell through a conduit while still retaining a seal with respect to an environment external to the electrolytic cell and external to the conduit.

[0028] FIG. 1 shows one non-limiting embodiment of a system **100** comprising an electrolytic cell **102** and a container **104** containing the electrolytic cell.

[0029] As described in more detail elsewhere herein, certain methods described herein may be performed in electrolytic cells and certain systems described herein may comprise an electrolytic cell. Electrolytic cells may be capable of performing and/or configured to perform one or more

redox reactions upon the input of electrical energy. The reactions that occur upon the input of such electrical energy may also be referred to as electrolytic reactions, and the process of operating an electrolytic cell to perform such reactions may be referred to as electrolysis. Operation of an electrolytic cell may comprise generating a voltage difference of one or more anodes present in the electrolytic cell with respect to one or more cathodes present therein, which may cause the anode(s) to exhibit a positive charge and the cathode(s) to exhibit a negative charge. The voltage difference may cause an oxidation reaction to occur at the anode and/or a reduction reaction to occur at the cathode.

[0030] In some instances, during electrolysis, the anode(s) and cathode(s) present in an electrolytic cell do not react during the redox reactions and so remain unconsumed by these redox reactions. It is also possible for either or both of the anode(s) and the cathode(s) to react during the redox reactions and/or to be consumed by the redox reactions. The redox reactions may comprise reducing metal cations present in the electrolyte to generate, as a desired product, elemental and/or alloyed metal (e.g., a metal having a zero oxidation state). The redox reactions may comprise oxidizing anion counter ions to form, as a byproduct, elemental gases.

[0031] FIG. 2 shows an electrolytic cell in more detail. The electrolytic cell shown in FIG. 2 comprises an anode **206** and a cathode **208**. It is contained within the container **204**. In some embodiments, an anode present in an electrolytic cell is disposed on a current collector and/or a cathode present in an electrolytic cell is disposed on a current collector (not shown).

[0032] In some embodiments, like the embodiments shown in FIGS. 1 and 2, an electrolytic cell is completely contained within a container. In other words, all portions of the electrolytic cell may be present in a volume enclosed by the container. It is also possible for an electrolytic cell to be partially contained within a container. In such embodiments, some portions of the electrolytic cell are present in a volume enclosed by the container and some portions of the electrolytic cell are present in a volume external to the container. As one example, in some embodiments, an electrolytic cell comprises a cathode and/or an anode that is partially contained within a container and further comprises a portion that extends out of the container. It is also possible for an electrolytic cell to comprise a voltage source, one or more leads, and/or one or more other components that are positioned outside a container in which the electrodes are partially or fully contained.

[0033] In some embodiments, an electrolytic cell further comprises an electrolyte, such as an electrolyte in which the electrodes are partially or fully immersed. Electrodes that are partially immersed in an electrolyte comprise one or more surfaces that are not in contact with the electrolyte, and electrodes that are fully immersed lack such surfaces. FIG. 3 shows one non-limiting embodiment of a system that comprises an electrolytic cell that comprises an electrolyte **310** in which the electrodes **306** and **308** are partially immersed.

[0034] In some embodiments, the electrolyte present in an electrolytic cell is contained in a further container positioned inside the container containing the electrolytic cell (also referred to herein as an “electrolyte container”). This further container may take the form of a crucible or other open-topped container, which may therefore allow fluidic connection between the electrolyte (and/or any other components contained in the electrolyte container) and an environment internal to the container. FIG. 4 shows one non-limiting example of a system having this design. In the system **400**, the electrolyte **410** is contained in the electrolyte container **412**.

[0035] As noted above, in some embodiments, a container for an electrolytic cell comprises a plurality of reversibly sealable ports. FIG. 5 shows one non-limiting example of such a container. In FIG. 5, the container **504** comprises the reversibly sealable ports **512A** and **512B**. When unsealed, the ports may allow for fluidic connection to be established between an environment contained by the container (e.g., an environment in which the electrolytic cell is positioned) and an environment in fluidic connection with and/or external to the container (e.g., an environment from which one or more species may be introduced into the environment contained by the container).

and/or into the electrolytic cell). When all such ports are sealed, in some embodiments, the environment contained by the container is not in fluidic connection with any environment in fluidic connection with and/or external to the container through such ports.

[0036] It should also be noted that some reversibly sealable ports, when sealed, may be sealed with respect to some species but not others. For instance, some reversibly sealable ports may be permeable to some gases (e.g., water vapor) but not others. In some embodiments, a reversibly sealable port is, when sealed, sealed with respect to water vapor and/or liquid water.

[0037] In some embodiments, a reversibly sealable port is, when sealed, sealed by a sealing material. In some such embodiments, a sealing material may cover a port and/or be positioned in a port in a manner that seals the port. A reversibly sealable port may comprise a sealing material, such as a sealing material that can be employed to reversibly seal (and/or unseal) the port. One non-limiting example of a sealing material is a rubber septum. Another non-limiting example is a membrane (e.g., a membrane that is selectively permeable to one or more species and/or gas, such as CO₂ and/or N₂).

[0038] As shown in FIG. 5, reversibly sealable ports may be positioned above an electrolytic cell and/or one or more components thereof (e.g., an electrolyte). Placement of reversibly sealable ports above an electrolyte may reduce or prevent contact between the electrolyte and the reversibly sealable ports. This may desirably reduce contamination of the reversibly sealable ports by the electrolyte.

[0039] In some embodiments, a container (and an electrolytic cell therein) is fluidically connected to an environment. For instance, in some embodiments, a container is immersed in and/or surrounded by an environment external thereto. FIG. 6 shows one non-limiting embodiment of a container positioned in such an environment, as it depicts the container 604 in fluidic connection with (via immersion in) the environment 616. It is also possible for a container to be in fluidic connection with an environment in which it is not immersed (not shown). As one example, in some embodiments, a conduit fluidically connects some or all of the external surface of a container to an environment. In some embodiments, a container is fluidically connected to an environment that one or more components of an electrolytic cell contained therein are not connected to. In such embodiments, some or all of the external surface of the electrolytic container may be fluidically connected to the environment and the container may seal the components of the electrolytic cell contained therein from the environment.

[0040] In some embodiments, an environment with which a container is in fluidic connection comprises a first species reactive with a second species present in the electrolytic cell during the performance of an electrolytic reaction. In some such embodiments, the container may be sealed with respect to that first species (e.g., it may maintain the environment internal to the container such that the second species makes up less than 5 wt %, less than 2 wt %, or less than 1 wt % of the gases therein). It is also possible, additionally or alternatively, for the container to be sealed with respect to that second species (e.g., it may maintain the environment with which the container is in fluidic connection such that the first species makes up less than 5 wt %, less than 2 wt %, or less than 1 wt % of the gases therein). When the container is sealed with respect to the second species, it may assist with trapping the second species. When the container is sealed with respect to both such species, it may thereby prevent reaction therebetween. In some embodiments, a container is sealed such that a pressure difference of at least 3 in H₂O can be maintained across one or more ports thereof.

[0041] FIG. 7 depicts one non-limiting example of a method. The method shown in FIG. 7 comprises the first optional step 718 of performing an electrolytic reaction in an electrolytic cell. As noted above, such a reaction may comprise performing a redox reaction (e.g., an oxidation reaction at an anode and/or a reduction reaction at a cathode), such as a redox reaction driven by a voltage difference established between an anode and a cathode.

[0042] The method shown in FIG. 7 further comprises several additional optional steps, which are

described in further detail below. Some or all such optional steps may be performed in combination with the optional step **718**. It should also be noted that some or all such optional steps may be performed concurrently with each other and/or with the step **718** and/or may be performed in a different order than shown in FIG. 7. As one example, and as shown in FIG. 7, the optional steps **720** and **722**, as described in further detail below, may be performed concurrently with each other. Additionally, it should be noted that some steps depicted as separate steps in FIG. 7 may be performed as part of a single step. For instance, the optional steps **720** and **722** may be performed as part of the optional step **718**.

[0043] The second optional step shown in FIG. 7 is the optional step **720**, which comprises reducing metallic ions to form a metal. As noted above, when performed, this step may be performed as part of the step **718**. In other words, an electrolytic reaction may be performed that comprises reducing metallic ions to form a metal. In some such embodiments, this reaction may occur at a cathode.

[0044] The third optional step shown in FIG. 7 is the optional step **722**, which comprises forming a byproduct. As noted above, when performed, this step may be performed as part of the step **718** and/or concurrently with the step **718**. In other words, an electrolytic reaction may be performed that comprises forming a byproduct (e.g., concurrently with reducing metallic ions to form a metal). In some embodiments, the formation of a byproduct and the reduction of metallic ions to form a metal each take the form of half reactions that together form a redox reaction. In some such embodiments, the formation of a byproduct occurs via an oxidation reaction at an anode (e.g., an oxidation of a counter ion to the metallic ion being reduced).

[0045] The fourth optional step shown in FIG. 7 is the optional step **724**. The optional step **724** comprises trapping the byproduct. The byproduct may be trapped in a variety of suitable manners, such as by the container (e.g., when the container is impermeable to the byproduct and the ports therein are sealed with respect to the byproduct). In some embodiments, a byproduct is trapped during one or more other steps described herein, such as during the performance of an electrolytic reaction.

[0046] The fifth optional step shown in FIG. 7 is the optional step **726**. The optional step **726** comprises depositing the byproduct on a component of the electrolytic cell. The deposition of the byproduct may be physical (e.g., the deposition of the byproduct may comprise a condensation of the byproduct on the component) and/or may comprise a chemical reaction between the byproduct and the component.

[0047] In some embodiments, deposition of a byproduct on a component of an electrolytic cell passivates the component. The passivation may comprise a reduction in the reactivity of the component, such as a reduction in the reactivity of the component with one or more species present in the electrolytic cell (e.g., in the electrolyte thereof) and/or contained in the container (e.g., in an environment contained therein, such as in a gaseous environment positioned above the electrolyte). Such a species may be a species that is present during the beginning of the electrolytic reaction and/or may be a species that is generated during the electrolytic reaction (e.g., a different byproduct, a reaction product of a different byproduct). The passivation may comprise forming a material that physically blocks access to the passivated component (e.g., the passivation may comprise a surface layer and/or a coating through which the relevant species cannot be and/or is not transported during the electrolytic reaction) and/or may comprise forming a new material via reaction with the passivated component that is non-reactive with one or more species present in the electrolytic cell and/or contained by the container. In some embodiments, passivation comprises the formation of a layer (e.g., a passivating layer) via either or both of these processes. Without wishing to be bound by any particular theory, it is believed that passivation of electrolytic cell components may improve the lifetime of such cells by reducing failure due to undesirable reactivity of such components.

[0048] The sixth optional step shown in FIG. 7 is the optional step **728**. The optional step **728**

comprises reacting the byproduct to form a reaction product of the byproduct. The reaction product may be formed after a single reaction between the byproduct and another species (e.g., another species contained by the container and/or present in the electrolytic cell) and/or may be formed after a series of reactions (e.g., in which the byproduct is reacted with a first species in a first reaction, the reaction product of that reaction is reacted with the first species or another species in a second reaction, and/or the reaction product of the second reaction is reacted with the first species or another species in a third reaction, etc.).

[0049] The seventh and eighth optional steps shown in FIG. 7 are the optional steps **730** and **734**. The optional step **730** comprises trapping the reaction product of the byproduct. This step may be performed substantially as described above with respect to the trapping of the byproduct. The optional step **732** comprises depositing the reaction product of the byproduct on a component of the electrolytic cell. This step may be performed substantially as described above with respect to the deposition of the byproduct on a component of the electrolytic cell. Similarly, it may passivate the component of the electrolytic cell substantially as described above with respect to the passivation of the component of the electrolytic cell by the byproduct.

[0050] The ninth optional step shown in FIG. 7 is the optional step **734**. The optional step **734** comprises depositing a species other than a byproduct or a reaction product of the byproduct on a component of the electrolytic cell. This step may be performed substantially as described above with respect to the deposition of the byproduct on a component of the electrolytic cell. Similarly, it may passivate the component of the electrolytic cell substantially as described above with respect to the passivation of the component of the electrolytic cell by the byproduct. The species other than the byproduct or the reaction product of a byproduct may be a species that does not undergo an electrolytic reaction but is otherwise present in the electrolytic cell, such as a species present in an electrolyte thereof and/or a gaseous atmosphere positioned above the electrolyte thereof.

[0051] In some embodiments, a method comprises performing the step **734** of the method shown in FIG. 7 without also performing an electrolytic reaction and/or without performing some or all of the other steps shown in FIG. 7.

[0052] FIG. 8 shows another exemplary method. The method shown in FIG. 8 may comprise some or all of the steps of the method shown in FIG. 7 and/or may comprise one or more further steps performed before and/or after such steps.

[0053] The first optional step shown in FIG. 8 is the step **834**, which comprises introducing a solid comprising metallic ions through an unsealed port. The unsealed port may be present in a system and/or in a container present in the system. The system may further comprise one or more additional ports (that may be sealed or unsealed during this step). During this step, the system and/or container may be heated. Without wishing to be bound by any particular theory, heating the system may reduce and/or prevent the introduction of liquid water into the container and/or the system during the introduction of the solid. As water may undesirably react with some electrolytic cell components, this may be desirable.

[0054] The second optional step shown in FIG. 8 is the step **836**, which comprises sealing the port. Sealing the port may desirably seal the container (e.g., if the container is otherwise sealed). This may allow for an electrolytic reaction to be performed in an electrolytic cell contained therein while the container is sealed. In some embodiments, the step **836** comprises sealing all of the ports present in a system and/or container.

[0055] The third optional step shown in FIG. 8 is the step **818**, which comprises performing an electrolytic reaction in the electrolytic cell. This step may comprise performing some or all of the steps of the method shown in FIG. 7. For instance, it may comprise forming a byproduct and/or trapping a byproduct, and/or reacting the byproduct to form a reaction product of the byproduct and/or trapping the reaction product of the byproduct. Similarly, it may comprise reducing metallic ions to form a metal. Such metal ions may originate from the solid comprising metallic ions and/or may be present in a molten salt electrolyte (e.g., a molten salt electrolyte formed upon melting

and/or dissolution of the solid comprising metallic ions).

[0056] As noted above, in some embodiments, the step **818** is performed while all of the ports in the system are sealed. As described elsewhere herein, this may assist with trapping byproducts (and/or reaction products thereof) in the electrolytic cell and/or of preventing the introduction of species from an environment in fluidic connection with the container into the electrolytic cell.

[0057] The fourth optional step shown in FIG. **8** is the step **840**. The step **840** comprises depositing a species trapped in the electrolytic cell on a component of the electrolytic cell. The deposited species may be a species generated in the electrolytic cell, such as a byproduct and/or a reaction product of the byproduct. Accordingly, the optional step **840** of the method shown in FIG. **8** may correspond to and/or comprise performing the optional steps **726** and/or **732** of the method shown in FIG. **7**. It is also possible for the trapped species to be a species introduced into the electrolytic cell (e.g., during introduction of the solid comprising metallic ions, as a contaminant). Accordingly, in such embodiments, the optional step **840** of the method shown in FIG. **7** may correspond to and/or comprise performing the optional step **734** of the method shown in FIG. **7**. In some embodiments, the deposited species passivates the component on which it is deposited.

[0058] The fifth optional step shown in FIG. **8** is the step **842**. The step **842** comprises unsealing a port. This step may be performed subsequent to the deposition of a species onto a component of an electrolytic cell and/or the passivation of such a component. This may allow for exposure of the passivated component to an environment with which the port is in fluidic connection (e.g., with which the container is also in fluidic connection) without appreciable reactivity thereof, even if the environment comprises one or more species that are reactive with the component when unpassivated.

[0059] In some embodiments, unsealing of a port may allow for the removal of a byproduct and/or a reaction product of a byproduct after passivation is complete and/or has proceeded to an acceptable level. Removing the byproduct and/or reaction product thereof at such a time point may be desirable if retaining the byproduct and/or reaction product of the byproduct in the system exhibits one or more disadvantages, such as being prone to further side reactions (e.g., with products of the electrolytic reaction, with the electrolyte). After such unsealing, the port may be resealed and the electrolytic reaction allowed to further proceed.

[0060] Additionally or alternatively, the sixth optional step shown in FIG. **8** may be performed. The sixth optional step shown in FIG. **8** is the step **844**. The step **844** comprises removing a metal through the unsealed port (e.g., the port unsealed in the step **842**, a port that is sealed after the step **842** and then resealed, or a different port). The metal may be a metal generated from the electrolytic reaction performed in the electrolytic cell (e.g., from reduction of metallic ions). In some embodiments, the system may be heated during such removal. This may be performed for substantially the same reasons described above with respect to the heating of the system during the introduction of the solid material comprising metallic ions.

[0061] The seventh optional step shown in FIG. **8** is the step **846**. The step **846** comprises cooling the reversibly sealable ports. This may be performed as the system is heated. Without wishing to be bound by any particular theory, this may desirably cause any gaseous water present during removal of the metal to condense on the reversibly sealable ports instead of being introduced into the electrolytic cell and/or container.

[0062] In some embodiments, one or more of the steps of a method described herein, such as a method comprising some or all of the steps shown in FIG. **7** and/or FIG. **8**, is performed while an electrolytic cell in which an electrolytic reaction is occurring is contained within a container. As noted above, this container may assist with trapping a byproduct (and/or a reaction product of a byproduct) generated during the electrolytic reaction and/or contain a component of the electrolytic cell onto which the byproduct (and/or the reaction product of the byproduct) is deposited.

[0063] A variety of suitable metallic ions may undergo electrolytic reactions in the electrolytic cells described herein to form a variety of suitable metals. In some embodiments, the metallic ions

comprise rare-earth metal ions, a rare-earth metal is formed upon reduction of metallic ions, and/or an alloy comprising a rare-earth metal is formed upon reduction of metallic ions. The “rare-earth metals,” as used herein, are the lanthanides, yttrium (Y), and scandium (Sc). The “lanthanides,” as used herein, are lanthanum (La), cerium (Ce), praseodymium (Pr), neodymium (Nd), promethium (Pm), samarium (Sm), europium (Eu), gadolinium (Gd), terbium (Tb), dysprosium (Dy), holmium (Ho), erbium (Er), thulium (Tm), ytterbium (Yb), and lutetium (Lu).

[0064] In some embodiments, the metallic ions comprise alkali metal ions, an alkali metal is formed upon reduction of metallic ions, and/or an alloy comprising an alkali metal is formed upon reduction of metallic ions. The “alkali metals,” as used herein, are lithium (Li), sodium (Na), potassium (K), rubidium (Rb), cesium (Cs), and francium (Fr).

[0065] In some embodiments, the metallic ions comprise alkaline earth metal ions, an alkaline earth metal is formed upon reduction of metallic ions, and/or an alloy comprising an alkaline earth metal is formed upon reduction of metallic ions. The “alkaline earth metals,” as used herein, are beryllium (Be), magnesium (Mg), calcium (Ca), strontium (Sr), barium (Ba), and radium (Ra).

[0066] In some embodiments, the metallic ions comprise transition metal ions, a transition metal is formed upon reduction of metallic ions, and/or an alloy comprising a transition metal is formed upon reduction of metallic ions. The “transition metals,” as used herein, are scandium (Sc), yttrium (Y), lanthanum (La), actinium (Ac), titanium (Ti), zirconium (Zr), hafnium (Hf), rutherfordium (Rf), vanadium (V), niobium (Nb), tantalum (Ta), dubnium (Db), chromium (Cr), molybdenum (Mo), tungsten (W), seaborgium (Sg), manganese (Mn), technetium (Tc), rhenium (Re), bohrium (Bh), iron (Fe), ruthenium (Ru), osmium (Os), hassium (Hs), cobalt (Co), rhodium (Rh), iridium (Ir), meitnerium (Mt), nickel (Ni), palladium (Pd), platinum (Pt), darmstadtium (Ds), copper (Cu), silver (Ag), gold (Au), roentgenium (Rg), zinc (Zn), cadmium (Cd), mercury (Hg), and copernicium (Cn).

[0067] In some embodiments, the metallic ions comprise aluminum (Al) ions, aluminum metal is formed upon reduction of metallic ions, and/or an alloy comprising aluminum is formed upon reduction of metallic ions.

[0068] A variety of byproducts may be generated during the electrolytic reactions described herein. As one example, in some embodiments, a byproduct comprises a halogen atom. The “halogens,” as used herein, are fluorine (F), chlorine (Cl), bromine (Br), iodine (I), astatine (At), and tennessine (Ts). The halogen atom may be bonded (e.g., covalently, ionically) to one or more other atoms. In one embodiment, the halogen may be present as a part of a halogen salt, such as in the form of a halide.

[0069] Byproducts generated during the electrolytic reactions described herein may be gaseous. In some embodiments, a byproduct both comprises a halogen and is gaseous. Such a byproduct may be a “halogen gas” and/or a “halide gas,” the latter of which refers to a gas comprising a cation (e.g., a hydrogen cation) and a halide. One non-limiting example of a suitable halogen gas is Cl.sub.2. It is also possible for a byproduct to comprise a gas comprising an oxygen atom and/or lacking halogen atoms, such as O.sub.2, CO, CO.sub.2, sulfur, and/or a sulfur-containing gas.

[0070] Byproducts generated during the electrolytic reactions may undergo a variety of types of further reactions in order to generate reaction products thereof. The byproducts may react with electrolytic cell components (e.g., electrodes, such as anodes and/or cathodes, current collectors, electrolytes) to form such reaction products and/or may react with one or more contaminants present in an electrolytic cell to form such reaction products (e.g., with a contaminant initially introduced into the electrolytic cell, with a contaminant entering the electrolytic cell through a reversibly scalable port). It is also possible for a byproduct to react with a species supplied to the electrolytic cell for the purpose of causing formation of the desired reaction product. For instance, a byproduct may react with oxygen gas supplied to an electrolytic cell for this purpose.

[0071] In some embodiments, a byproduct undergoes a Boudouard reaction to form a reaction product. Without wishing to be bound by any particular theory, a Boudouard reaction may be

described by the following equation:

$2\text{CO} \rightarrow \text{C} + \text{CO}_2$ (Equation 1).

[0072] In such embodiments, the byproduct and/or reaction product may be CO and/or the component being passivated may be passivated by the formation of a carbon layer thereon created by the Boudouard reaction. It is also possible for the byproduct and/or reaction product to comprise CO₂, for the Boudouard reaction to result in the creation of a further CO reaction product, and for the CO to passivate a component of an electrolytic cell and/or undergo one or more further reactions to yield a reaction product that passivates a component of an electrolytic cell (e.g., the CO may undergo a further Boudouard reaction to reform carbon that passivates an electrolytic cell component).

[0073] In some embodiments, the carbon undergoing the Boudouard reaction may take the form of carbon dust, and the Boudouard reaction may desirably result in the removal of carbon dust from the electrolytic cell. It is also possible for reactions in an electrolytic cell to occur that include the reaction of carbon dust via the Boudouard reaction to create CO that is then redeposited as carbon on a component of an electrolytic cell. This carbon may passivate that component of the electrolytic cell. In other words, Boudouard reactions may occur in an electrolytic cell that result in the transport of carbon from carbon dust to a passivating layer disposed on a component of the electrolytic cell. These reactions may be accomplished by the formation of temperature gradients in the electrolytic cell that favor the reaction of carbon dust via the Boudouard reaction and the deposition of carbon via the Boudouard reaction on an electrolytic cell component.

[0074] In some embodiments, a species other than a byproduct or a reaction product of a byproduct undergoes a Boudouard reaction. For instance, O₂ that is not a byproduct or a reaction product of a byproduct may undergo one or more reactions to form CO₂ and/or CO, which may undergo the Boudouard reaction. It should also be noted that some species that are not byproducts or reaction products of byproducts may have a composition described above as being a suitable composition for a byproduct and/or a composition described above as being a suitable composition for a reaction product of a byproduct.

[0075] As noted above, in some embodiments, an electrolytic cell comprises an electrolyte. In some embodiments, an electrolyte comprises a molten salt. It is also possible for the electrolyte to be a molten salt electrolyte (e.g., it may comprise, consist of, and/or consist essentially of a molten salt). A “molten salt,” as used herein, is a liquid-phase salt. In some embodiments, a molten salt may comprise, consist essentially of, and/or consist of a salt that is present at a temperature above its melting point. It is also possible for a molten salt to further comprise a dissolved metal and/or a dissolved gas (e.g., a dissolved metal and/or a dissolved gas that is produced by a reaction, such as a redox reaction, occurring in an electrolytic cell present in the system). A molten salt is not the same as a solubilized salt (which is a salt that has been solubilized into its constituent ions within a solvent). In some embodiments, the molten salt is a salt that is in a solid phase when at a temperature of 25° C. and a pressure of 1 atmosphere but that melts to form a liquid phase when heated to or above its melting point. In some embodiments, the molten salt is a salt that is formed upon the melting of a solid comprising metallic ions initially introduced into an electrolytic cell.

[0076] In some embodiments, a method comprises performing a reaction that passivates a component of an electrolytic cell and/or an electrolytic cell comprises a passivated component. Non-limiting examples of components that may be so passivated include electrodes (e.g., anodes, cathodes), current collectors, electrolyte containers. In some embodiments, a component that is passivated comprises carbon, a metal (e.g., niobium (Nb), a refractory metal), an alloy (e.g., steel, Hastelloy, an iron alloy), and/or a ceramic. In some embodiments, a component that is passivated comprises stainless steel and/or a low-carbon steel. The above-described materials may themselves be passivated (e.g., a component that is passivated may comprise an alloy whose reactivity is reduced due to passivation) and/or may be present in a component that further comprises another

material that is passivated (e.g., without, themselves, being passivated).

[0077] In some embodiments, a method described herein causes the formation of a passivating layer on an electrolytic cell component (i.e., a layer that passivates the electrolytic cell component). Passivating layers may be relatively stable or may degrade upon exposure to one or more conditions. In some embodiments, a passivating layer is stable during electrolytic reactions occurring in the electrolytic cell (in addition to being stable other under conditions or despite degrading under other conditions). For instance, a passivating layer may be unreactive with any species generated at the passivated component during such electrolytic reactions (e.g., any gases generated at the passivated component) and/or to which the passivated component is exposed during such electrolytic reactions.

[0078] As one example an unstable passivating layer, in some embodiments, a passivating layer is unstable upon exposure to species present in an environment with which a container is in fluidic connection and/or upon exposure to species generated during an electrolytic reaction. In such embodiments, it may be desirable to limit the exposure of the passivating layer to conditions that cause it to degrade. This may be accomplished by, for instance, keeping the container in which the electrolytic cell is positioned sealed when materials are not being added thereto and/or removed therefrom.

[0079] Passivating layers described herein may have a variety of suitable compositions. In some embodiments, passivating layers comprise carbon. Without wishing to be bound by any particular theory, it is believed that carbon-containing passivating layers may be unstable upon exposure to air. As noted above, in such instances, it may be desirable to limit the exposure of such passivating layers to air. It is also possible for passivating layers to comprise ceramics and/or salts, such as halide salts (e.g., ferric chloride).

[0080] Passivating layers may have a variety of suitable thicknesses. In some embodiments, a passivating layer has an average thickness of less than or equal to 3 mm, less than or equal to 2 mm, less than or equal to 1 mm, less than or equal to 0.75 mm, less than or equal to 0.5 mm, or less than or equal to 0.2 mm. In some embodiments, a passivating layer has an average thickness of greater than or equal to 0.1 mm, greater than or equal to 0.2 mm, greater than or equal to 0.5 mm, greater than or equal to 0.75 mm, greater than or equal to 1 mm, or greater than or equal to 2 mm. Combinations of the above-referenced ranges are also possible (e.g., less than or equal to 3 mm and greater than or equal to 0.1 mm). Other ranges are also possible.

[0081] In some embodiments, a method as described herein causes an electrolytic cell to have a relatively low concentration of free carbon dust (e.g., particulate carbon that is not spatially fixed by any electrolytic cell component) and/or a relatively low rate of generation of free carbon dust. Similarly, some systems described herein may include a relatively low amount of free carbon dust. Without wishing to be bound by any particular theory, it is believed that free carbon dust may be indicative of undesirable side reactions occurring at electrolytic cell components comprising carbon and/or may itself undergo undesirable side reactions with electrolytic cell components (e.g., with the electrolyte, one or more components thereof, and/or metal generated therein). Such side reactions may comprise reaction with one or more contaminants present in the electrolytic cell, such as water and/or air. In some embodiments, free carbon dust does not passivate any electrolytic cell component.

[0082] In some embodiments, the amount of free carbon dust present in an electrolytic cell is less than or equal to 5 wt %, less than or equal to 3 wt %, less than or equal to 2 wt %, less than or equal to 1 wt %, less than or equal to 0.75 wt %, less than or equal to 0.5 wt %, or less than or equal to 0.2 wt % of an amount of electrolyte in the electrolytic cell. In some embodiments, the amount of free carbon dust present in an electrolytic cell is greater than or equal to 0.1 wt %, greater than or equal to 0.2 wt %, greater than or equal to 0.5 wt %, greater than or equal to 0.75 wt %, greater than or equal to 1 wt %, greater than or equal to 2 wt %, or greater than or equal to 3 wt % of an amount of electrolyte in the electrolytic cell. Combinations of the above-referenced ranges

are also possible (e.g., less than or equal to 5 wt % and greater than or equal to 0.1 wt %). Other ranges are also possible.

[0083] In some embodiments, free carbon dust is generated at a rate of less than or equal to 20%, less than or equal to 15%, less than or equal to 10%, less than or equal to 7.5%, less than or equal to 5%, or less than or equal to 2% of a rate at which metal is generated in the electrolytic cell. In some embodiments, free carbon dust is generated at a rate of greater than or equal to 1%, greater than or equal to 2%, greater than or equal to 5%, greater than or equal to 7.5%, greater than or equal to 10%, or greater than or equal to 15% of a rate at which metal is generated in the electrolytic cell. Combinations of the above-referenced ranges are also possible (e.g., less than or equal to 20% and greater than or equal to 1%). Other ranges are also possible.

[0084] As also noted above, in some embodiments, an electrolytic cell is positioned in a container that is fluidically connected to an environment, such as environment comprising a first species reactive with a second species present in the electrolytic cell during the performance of an electrolytic reaction. Non-limiting examples of such first species include water and oxygen.

[0085] Second species (i.e., species reactive with a first species as described above) may be generated during an electrolytic reaction (e.g., a byproduct produced thereby, a reaction product of such a byproduct), may be present in a component of the electrolytic cell (e.g., it may be positioned in an electrode, such as an anode and/or a cathode, and/or may be positioned in a current collector), may be a reagent consumed during the electrolytic reaction (e.g., it may be a metallic ion or a counter ion thereto), or may be present in an electrolyte present during the electrolytic reaction. It is also possible for two or more types of second species to be present in an electrolytic cell. For instance, an electrolytic cell may comprise both a byproduct and an electrode that are reactive with a first species as described herein. Non-limiting examples of second species include carbon and metals.

[0086] In some embodiments, a first species (e.g., a byproduct, a reaction product of a byproduct, a species other than a byproduct or a reaction product of a byproduct) present in an environment is reactive with a second species present in an electrolytic cell in a manner that generates a third species. The third species may be reactive with a component of an electrolytic cell and/or a species present in a component of an electrolytic cell, such as an electrode therein (e.g., an anode therein, a cathode therein, a current collector therein), a current collector therein, and/or an electrolyte container therein. The component of the electrolytic cell and/or species therein may comprise carbon and/or a metal. In some embodiments, the reaction between the third species and the component of the electrolytic cell (and/or species present therein) may be undesirable. For instance, it may result in undesirable corrosion and/or etching of the component and/or species. In such embodiments, it may be desirable to reduce generation of the third species, such as by limiting exposure of the second species to the first species (e.g., by sealing the container with respect to the first species).

[0087] The temperature of an electrolyte in an electrolytic cell during an electrolytic reaction may be selected as desired. In some embodiments, the temperature of an electrolyte in an electrolytic cell during an electrolytic reaction is greater than or equal to 100° C., greater than or equal to 200° C., greater than or equal to 500° C., greater than or equal to 750° C., greater than or equal to 1000° C., or greater than or equal to 1200° C. In some embodiments, the temperature of an electrolyte in an electrolytic cell during an electrolytic reaction is less than or equal to 1400° C., less than or equal to 1200° C., less than or equal to 1000° C., less than or equal to 750° C., less than or equal to 500° C., or less than or equal to 200° C. Combinations of the above-referenced ranges are also possible (e.g., greater than or equal to 200° C. and less than or equal to 1400° C.). Other ranges are also possible.

[0088] As noted above, in some embodiments, a system comprises a plurality of reversibly scalable ports, such as reversibly scalable ports comprising scaling materials. In some embodiments, a system further comprises a cooling subsystem in fluidic connection with the sealing material. The

cooling subsystem may assist with maintaining the temperature of the sealing material in a desirable range, such as a range that promotes effective sealing and/or promotes or reduces condensation of gases contained by the container on the sealing material and/or the port.

[0089] In some embodiments, a cooling subsystem maintains, is configured to maintain, and/or is capable of maintaining a sealing material at a temperature of greater than or equal to 25° C., greater than or equal to 50° C., greater than or equal to 75° C., greater than or equal to 100° C., greater than or equal to 150° C., greater than or equal to 200° C., greater than or equal to 250° C., greater than or equal to 300° C., greater than or equal to 350° C., greater than or equal to 400° C., greater than or equal to 450° C., greater than or equal to 500° C., or greater than or equal to 550° C. In some embodiments, a cooling subsystem maintains, is configured to maintain, and/or is capable of maintaining a sealing material at a temperature of less than or equal to 600° C., less than or equal to 550° C., less than or equal to 500° C., less than or equal to 450° C., less than or equal to 400° C., less than or equal to 350° C., less than or equal to 300° C., less than or equal to 250° C., less than or equal to 200° C., less than or equal to 150° C., less than or equal to 100° C., less than or equal to 75° C., or less than or equal to 50° C. Combinations of the above-referenced ranges are also possible (e.g., greater than or equal to 25° C. and less than or equal to 600° C.). Other ranges are also possible.

[0090] As noted above, in some embodiments, a system is heated during the introduction of a solid material comprising metallic ions into a system and/or the removal of metal from a system. In some embodiments, the system is heated to a temperature of greater than or equal to 25° C., greater than or equal to 50° C., greater than or equal to 75° C., greater than or equal to 100° C., greater than or equal to 200° C., greater than or equal to 500° C., greater than or equal to 750° C., or greater than or equal to 1000° C. during this introduction and/or removal. In some embodiments, the system is heated to a temperature of less than or equal to 1100° C., less than or equal to 1000° C., less than or equal to 750° C., less than or equal to 500° C., less than or equal to 200° C., less than or equal to 100° C., less than or equal to 75° C., or less than or equal to 50° C. during this introduction and/or removal. Combinations of the above-referenced ranges are also possible (e.g., greater than or equal to 25° C. and less than or equal to 1100° C.). Other ranges are also possible.

[0091] The introduction of a solid material comprising metallic ions into a system and the removal of metal from a system may be performed over a variety of suitable time periods. In some embodiments, such introduction and/or removal may take place over a period of time of less than or equal to 24 hours, less than or equal to 18 hours, less than or equal to 12 hours, less than or equal to 6 hours, less than or equal to 3 hours, less than or equal to 2 hours, or less than or equal to 1 hour. In some embodiments, such introduction and/or removal may take place over a period of time of greater than or equal to 30 minutes, greater than or equal to 1 hour, greater than or equal to 2 hours, greater than or equal to 3 hours, greater than or equal to 6 hours, greater than or equal to 12 hours, or greater than or equal to 18 hours. Combinations of the above-referenced ranges are also possible (e.g., less than or equal to 24 hours and greater than or equal to 30 minutes). Other ranges are also possible.

[0092] In some embodiments, a system comprises an oxygen lance. The oxygen lance may be capable of supplying oxygen to an interior of a container and/or to an electrolytic cell contained in the container. It is also possible for the oxygen lance to be configured to do so. In some embodiments, the oxygen is supplied without also supplying other species and/or other reactive species in an appreciable amount. For instance, in some embodiments, the oxygen may be supplied from an oxygen tank and/or in the presence of an inert gas. As noted above, in some embodiments, it may be desirable to form a reaction product of a byproduct by reaction of the byproduct with oxygen. The oxygen lance may be employed to supply oxygen for this purpose.

Example 1

[0093] This Example depicts a system suitable for performing some or all of the methods described herein. This system is shown schematically in FIG. 9. The system shown in FIG. 9 includes an

electrolytic cell comprising an anode and a cathode positioned in a container. The container further comprises a reversibly sealable port that gas may flow out of and into a gas treatment center. As shown in FIG. 9, gases generated at the anode during operation of the electrolytic cell may deposit and/or react to create a passivating (e.g., protective) layer at the interface between the anode and an environment contained by the container.

Example 2

[0094] This Example describes the performance of electrolytic reactions in sealed and unsealed electrolytic cells.

[0095] Electrolytic reactions were performed in an unsealed electrolytic cell comprising a tungsten cathode, graphite anodes, and stainless steel current collectors to produce Nd metal. Subsequent to the performance of the electrolytic reactions, the anode current collectors were removed from the electrolytic cell while it was still heated. These current collectors were observed to have corrosion and scale disposed thereon.

[0096] Otherwise equivalent electrolytic reactions were performed in an electrolytic cell otherwise equivalent to the unsealed electrolytic cell except that it was sealed and held under a vacuum of 0.5 in H.sub.2O. Subsequent to the performance of the electrolytic reactions, the anode current collectors were removed from the electrolytic cell while it was still heated. These current collectors were observed to lack corrosion and scale. Instead, a dark layer of fine carbon was disposed thereon. Underneath the dark layer of fine carbon, the anode current collectors had black coloration.

[0097] Additionally, after the performance of the electrolytic reaction, the interior of the electrolytic cell was inspected. During this inspection, finely dispersed carbon was observed throughout the electrolytic cell, leading to a conclusion that the Boudouard reaction had occurred. It is believed that the dark layer of fine carbon protected the anode current collectors at the temperatures present during the electrolytic reaction and upon their removal from the electrolytic cell, thereby preventing oxygen corrosion of these current collectors.

[0098] While several embodiments of the present invention have been described and illustrated herein, those of ordinary skill in the art will readily envision a variety of other means and/or structures for performing the functions and/or obtaining the results and/or one or more of the advantages described herein, and each of such variations and/or modifications is deemed to be within the scope of the present invention. More generally, those skilled in the art will readily appreciate that all parameters, dimensions, materials, and configurations described herein are meant to be exemplary and that the actual parameters, dimensions, materials, and/or configurations will depend upon the specific application or applications for which the teachings of the present invention is/are used.

[0099] Those skilled in the art will recognize, or be able to ascertain using no more than routine experimentation, many equivalents to the specific embodiments of the invention described herein. It is, therefore, to be understood that the foregoing embodiments are presented by way of example only and that, within the scope of the appended claims and equivalents thereto, the invention may be practiced otherwise than as specifically described and claimed. The present invention is directed to each individual feature, system, article, material, and/or method described herein. In addition, any combination of two or more such features, systems, articles, materials, and/or methods, if such features, systems, articles, materials, and/or methods are not mutually inconsistent, is included within the scope of the present invention.

[0100] The indefinite articles “a” and “an,” as used herein in the specification and in the claims, unless clearly indicated to the contrary, should be understood to mean “at least one.”

[0101] The phrase “and/or,” as used herein in the specification and in the claims, should be understood to mean “either or both” of the elements so conjoined, i.e., elements that are conjunctively present in some cases and disjunctively present in other cases. Other elements may optionally be present other than the elements specifically identified by the “and/or” clause, whether

related or unrelated to those elements specifically identified unless clearly indicated to the contrary. Thus, as a non-limiting example, a reference to “A and/or B,” when used in conjunction with open-ended language such as “comprising” can refer, in one embodiment, to A without B (optionally including elements other than B); in another embodiment, to B without A (optionally including elements other than A); in yet another embodiment, to both A and B (optionally including other elements); etc.

[0102] As used herein in the specification and in the claims, “or” should be understood to have the same meaning as “and/or” as defined above. For example, when separating items in a list, “or” or “and/or” shall be interpreted as being inclusive, i.e., the inclusion of at least one, but also including more than one, of a number or list of elements, and, optionally, additional unlisted items. Only terms clearly indicated to the contrary, such as “only one of” or “exactly one of,” or, when used in the claims, “consisting of,” will refer to the inclusion of exactly one element of a number or list of elements. In general, the term “or” as used herein shall only be interpreted as indicating exclusive alternatives (i.e. “one or the other but not both”) when preceded by terms of exclusivity, such as “either,” “one of,” “only one of,” or “exactly one of.” “Consisting essentially of,” when used in the claims, shall have its ordinary meaning as used in the field of patent law.

[0103] As used herein in the specification and in the claims, the phrase “at least one,” in reference to a list of one or more elements, should be understood to mean at least one element selected from any one or more of the elements in the list of elements, but not necessarily including at least one of each and every element specifically listed within the list of elements and not excluding any combinations of elements in the list of elements. This definition also allows that elements may optionally be present other than the elements specifically identified within the list of elements to which the phrase “at least one” refers, whether related or unrelated to those elements specifically identified. Thus, as a non-limiting example, “at least one of A and B” (or, equivalently, “at least one of A or B,” or, equivalently “at least one of A and/or B”) can refer, in one embodiment, to at least one, optionally including more than one, A, with no B present (and optionally including elements other than B); in another embodiment, to at least one, optionally including more than one, B, with no A present (and optionally including elements other than A); in yet another embodiment, to at least one, optionally including more than one, A, and at least one, optionally including more than one, B (and optionally including other elements); etc.

[0104] As used herein, “wt %” is an abbreviation of weight percentage. As used herein, “at %” is an abbreviation of atomic percentage.

[0105] Some embodiments may be embodied as a method, of which various examples have been described. The acts performed as part of the methods may be ordered in any suitable way. Accordingly, embodiments may be constructed in which acts are performed in an order different than illustrated, which may include different (e.g., more or less) acts than those that are described, and/or that may involve performing some acts simultaneously, even though the acts are shown as being performed sequentially in the embodiments specifically described above.

[0106] Use of ordinal terms such as “first,” “second,” “third,” etc., in the claims to modify a claim element does not by itself connote any priority, precedence, or order of one claim element over another or the temporal order in which acts of a method are performed, but are used merely as labels to distinguish one claim element having a certain name from another element having a same name (but for use of the ordinal term) to distinguish the claim elements.

[0107] In the claims, as well as in the specification above, all transitional phrases such as “comprising,” “including,” “carrying,” “having,” “containing,” “involving,” “holding,” and the like are to be understood to be open-ended, i.e., to mean including but not limited to. Only the transitional phrases “consisting of” and “consisting essentially of” shall be closed or semi-closed transitional phrases, respectively, as set forth in the United States Patent Office Manual of Patent Examining Procedures. Section 2111.03.

Claims

1. A method of passivating a component of an electrolytic cell, comprising: performing an electrolytic reaction in the electrolytic cell, wherein the electrolytic reaction comprises reducing metallic ions to form a metal and comprises forming a byproduct; and depositing the byproduct and/or a reaction product of the byproduct on the component, thereby passivating the component.
 2. A method, comprising: performing an electrolytic reaction in an electrolytic cell, wherein the electrolytic reaction comprises reducing metallic ions to form a metal and forming a byproduct, and wherein, during the electrolytic reaction, the electrolytic cell traps the byproduct and/or a reaction product of the byproduct.
 3. A method, comprising: performing an electrolytic reaction in an electrolytic cell, wherein: the electrolytic reaction comprises reducing metallic ions to form a metal, during the electrolytic reaction, the electrolytic cell is contained in a container, the container is fluidically connected to an environment, the environment comprises a first species reactive with a second species present in the electrolytic cell during the electrolytic reaction, and during the electrolytic reaction, the container is sealed with respect to the first species.
 - 4-5. (canceled)
 6. A method as in claim 1, wherein the metal is a rare-earth metal.
 7. A method as in claim 1, wherein the metal is an alkali metal.
 8. A method as in claim 1, wherein the metal is an alkaline earth metal.
 9. A method as in claim 1, wherein the metal is a transition metal.
 10. A method as in claim 1, wherein the metal is Al.
 11. (canceled)
 12. A method as in claim 1, wherein the electrolytic cell comprises an electrolyte that is a molten salt electrolyte.
 13. A method as in claim 1, wherein the method comprises depositing the byproduct on the component, thereby passivating the component.
 14. A method as in claim 1, wherein the method comprises depositing the reaction product of the byproduct on the component, thereby passivating the component.
 15. A method as in claim 1, wherein, during the electrolytic reaction, the electrolytic cell traps the byproduct and/or the reaction product of the byproduct.
 - 16-29. (canceled)
 30. A method as in claim 1, wherein the component comprises carbon.
 31. A method as in claim 1, wherein the component comprises a metal.
 32. A method as in claim 1, wherein the component is an electrode.
 - 33-34. (canceled)
 35. A method as in claim 1, wherein the component is a current collector.
 36. A method as in claim 1, wherein the component is an electrolyte container.
 - 37-50. (canceled)
 51. A method as in claim 1, wherein the reaction product of the byproduct is formed via a Boudouard reaction.
 52. A method as in claim 1, wherein the reaction product of the byproduct is formed via a reaction of the byproduct with the component.
 - 53-57. (canceled)
 58. A method as in claim 1, wherein the reaction product of the byproduct is formed via a reaction of the byproduct with oxygen.
 - 59-68. (canceled)
-